

Zhulong Xue

Curriculum Vitae

(Aug. 2026)

Personal Details

Name: Zhulong Xue

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Address: Academy of Mathematics and Systems Science, Chinese Academy of Sciences, 100190, Beijing, China

Research Interests

Analysis and partial differential equations

Employment

Jul. 2025 — present **Post-doc Researcher**, Chinese Academy of Sciences
Mentor: Quoc-Hung Nguyen

Education

Sep. 2020—Jun. 2025 **Beijing Normal University**, PhD in mathematics **Supervisor:** Liutang Xue
Sep. 2016—Jun. 2020 **Chang'an University**, BS in mathematics

Research Stays

Oct. 2023 — Oct. 2024 **Visiting PhD student**, University of Seville, Spain
Supervisor: Francisco Gancedo
Funded by China Scholarship Council
Sep. 2024 — Oct. 2024 **Visiting PhD student**, University of Granada, Spain
Host: Claudia García

Publications and Preprints

1. Zhulong Xue and Weicheng Zhan. Uniformly rotating vortex patches with arbitrarily many genuine holes. arXiv:2608.19627.
2. Changhui Tan, Liutang Xue, and Zhulong Xue. *A revisit of patch solutions for the 2D Loglog-Euler type equation*. arXiv:2510.18759.
3. Taoufik Hmidi, Liutang Xue, and Zhulong Xue. *Doubly connected V-states in geophysical models: a general framework*. Calculus of Variations and Partial Differential Equations **65** (2026), Paper No. 137.
4. Qianyun Miao, Changhui Tan, Liutang Xue, and Zhulong Xue. *Local regularity and finite-time singularity for a class of generalized SQG patches on the half-plane*. arXiv:2410.19273.
5. Taoufik Hmidi, Liutang Xue, and Zhulong Xue. *Unified theory on V-states structures for active scalar equations*. Advances in Mathematics **486** (2026), Paper No. 110750.
6. Taoufik Hmidi, Liutang Xue, and Zhulong Xue. *Emergence of time periodic solutions for the generalized surface quasi-geostrophic equation in the disc*. Journal of Functional Analysis **285** (2023), no. 10, Paper No. 110142.

Invited Talks

04/06/2026 Singularities, non-uniqueness and complex dynamics of incompressible fluids, Bath, UK
16/10/2024 PhD seminar, IMUS, University of Seville, Spain
21/03/2024 Ecuaciones Diferenciales seminar, IMAG, University of Granada, Granada, Spain
13/11/2023 Fluid Conversations seminar, IMUS, University of Seville, Spain

Conferences (Selected)

Aug. 2026 *Advances in Mathematical Fluid Dynamics 2026*, Peking University, China
Aug. 2026 *Workshop on Nonlinear PDEs in Fluid Mechanics*, Chinese Academy of Sciences, China
Jun. 2026 *Singularities, non-uniqueness and complex dynamics of incompressible fluids*, Bath, UK
May 2026 *Workshop: Analysis of PDEs in Dalian*, Dalian University of Technology, China
Jul. 2025 *Mathematical theory on Navier-Stokes equation*, Peking University, China
Jun. 2025 *Advances in Mathematical Fluid Dynamics*, Peking University, China
Nov. 2024 *Mixing, enhanced dissipation and stability effects in fluid dynamics*, Kunming, China
Jul. 2024 *Alhambra PDE days*, IMAG, University of Granada, Spain
Jul. 2024 *9th European Congress of Mathematics*, University of Seville, Spain

Funding

Jul. 2026 National Postdoctoral Researcher Funding Program (Grade B), China
Jul. 2026 General Program of China Postdoctoral Science Foundation, China